

	$\frac{1}{r^2}$
6(a)(i)	It is a region of space where a non-contact force acts on a particle by virtue of its particular property.
6(a)(ii)	It may be experiencing a magnetic force instead due to the presence of a magnetic field instead if the charge is moving. Secondly, the charge may

	experience a gravitational force since it has mass.
6(b)(i)	The two regions of x are inside the two spheres where electric field strength is zero as the charges on them will repel each other till all the charges are on the outer surface of the spheres. This result in the electric field strength to be zero inside the sphere.
6(b)(ii)1.	Sphere A : 4 cm, sphere B: 2 cm
6(b)(ii)2.	The spheres have charges of the opposite sign as the resultant E is always positive suggesting that its direction is in the same direction along the line joining the centres of the two spheres.
6(b)(iii)1 .	Energy gained = Work done on the nucleus = area under the $F - x$ graph = $q \times$ area under the $E - x$ graph = $(+3e)30(0.4 \times 10^5 \times 0.5 \times 10^{-2})$ = $2.88 \times 10^{-15} \text{ J}$
6(b)(iii)2 .	At $x = 25.0 \text{ cm}$, $E = 3.2 \times 10^5 \text{ V m}^{-1}$ From Newton's 2nd law, acceleration, $a = \frac{F}{m}$ $= \frac{qE}{m}$ $= \frac{(3e)(3.2 \times 10^5)}{7m_p}$ $= 1.3 \times 10^{13} \text{ m s}^{-2}$
6(b)(iii)3 .	As in 6(b)(iii)2 , the acceleration of the nucleus is proportional to the value of E . Thus, as x increases, the acceleration of the nucleus decreases as E decreases to a minimum value at about $x = 18.5 \text{ cm}$. After this point, the acceleration starts to increase again, in the same direction, until it reaches the same maximum value as at the start at $x = 28.0 \text{ cm}$.
7(c)(i)	The resistance of a wire is the ratio of the potential difference across it to the